

argument to *polyval*, the polynomial is evaluated at each of those points. Let us now redefine *r* as a square matrix and see the results:

```
octave -3.2.2:108 > r = [1 2 3 4];
octave -3.2.2:109 > polyval(c, r)
ans =
```

```
14 41
98 197
```

Polynomial curve fitting

The idea of polynomial curve fitting is that, given a set of input and output vectors, the best curve passing through the given points needs to be fitted. This is usually done by minimising the least squares distance between given points and the curve. For more background on this subject, please see the Wolfram Mathworld article given under *Resources*.

The *polyfit* function is used to find the best-fit curve for a set of input and output vectors. I shall illustrate this with an example from the MATLAB documentation of the same function (apologies for the digression, but this also illustrates the compatibility between MATLAB and Octave, a topic that we shall discuss in more detail in a later article):

```
octave -3.2.2:110 > x = [1 2 3 4 5];
y = [5 5.43 1.128 290.74 9.8 4];
```

```
octave -3.2.2:114 > p = polyfit(x, y, 3)
p =
```

```
-0.19167 31.58214 -60.32619 35.34000
```

```
octave -3.2.2:115 > x2 = 1:1:5;
octave -3.2.2:116 > y2 = polyval(p, x2);
octave -3.2.2:117 > plot(x, y, 'o', x2, y2)
octave -3.2.2:118 > grid on
```

You should now see a plot similar to the one shown in Figure 1. Okay, so what we have done here is obtained the coefficients *p* of a best-fit curve between *x* and *y* with *polyfit*. Next, we used the obtained coefficients to get the output values *y2* for the same *x* values over

a finer range *x2*, and then plotted the results along with our earlier *x,y* (indicated by the circles). As you can see from the plot, they are almost on the obtained curve.

By default, *polyfit* just returns the coefficients of the polynomial. However, if you also give a second return argument, like [*p*, *s*] = *polyfit*(*x*, *y*, 3), then it returns a structure of values containing the following fields:

- **R:** The Cholesky factor of the Vandermonde matrix used to compute the polynomial coefficients.
- **X:** The Vandermonde matrix used to compute the polynomial coefficients.
- **df:** The degrees of freedom.
- **normr:** The norm of the residuals.
- **yf:** The values of the polynomial for each value of *x*.

Other features: There are also various other functions to work with polynomials, such as *polyderiv* and *polyinteg* that return the coefficients of the derivative and the integral polynomial of the function, respectively.

Miscellaneous features

Here are some other facilities available in Octave that you can explore on your own.

Plotting: Briefly used earlier, Octave has got good facilities for plotting data in two and three dimensions. You can also interact

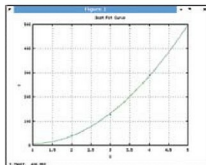


Figure 1: A best-fit curve

with *gnuplot* from within Octave, which is a very good feature to have.

Sets: Octave has some support for working with sets. A *set* is created with the *create_set* function. Set operations—such as *union*, *intersection*, *ismember*—are also supported in Octave.

Control system toolbox: The control system toolbox looks very well developed, though I have not explored or studied it in depth.

Image and audio processing: Octave has got some support for loading, display and some basic operations for images. There is also some support for audio processing.

Looking ahead

In the next article in this series, we will learn to write Octave scripts and take a step towards using Octave as a programming language rather than as a command interpreter, as we have been doing so far. See you next time! 

Resources

- GNU Octave: www.gnu.org/software/octave/
- GNU Octave Documentation: www.gnu.org/software/octave/docs.html
- GNU Octave Function reference: octave.sourceforge.net/doc/index.html
- Least Squares Fitting: mathworld.wolfram.com/LeastSquaresFitting.html

Acknowledgements

Considering the heavy usage of Math formulas, LaTeX was an automatic choice to write this article and thanks to the *The Not So Short Introduction to LaTeX* (<http://ctan.tug.org/tex-archive/info/short/english/short.pdf>) for a nice treatment of how to write Math well in LaTeX.

By: Amit Saha

The author blogs at <http://amitsaha.wordpress.com> and he welcomes queries and suggestions at amitsaha.in@gmail.com